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```
imprimir_tablero :: [[Int]] -> IO()
imprimir_tablero [] = putStrLn ("-----")
imprimir_tablero (x:xs) = do
    putStrLn ( show(x!!0) ++ " " ++ show(x!!1) ++ " " ++ show(x!!2) ++ " " ++ show(x!!3) ++ " "
    ++ show(x!!4) ++ " " ++ show(x!!5) ++ " " ++ show(x!!6) ++ " " ++ show(x!!7) ++ " " ++
    show(x!!8))
    imprimir_tablero (xs)
```

```
fila :: [Int] -> Bool
fila [] = True
fila (x:xs) = x /= 0 && (fila xs)
```

```
movimiento_col :: Int -> Int -> [Int] -> [Int]
movimiento_col x valor lista = (take x lista) ++ [valor] ++ (drop (x+1) lista)
```

```
movimiento :: Int -> Int -> Int -> [[Int]] -> [[Int]]
movimiento x y valor tablero = (take y tablero) ++ [movimiento_col x valor (tablero !! y)] ++
(drop (y+1) tablero)
```

```
m_cuadrados :: Int -> Int -> Int -> [[Int]] -> Bool
m_cuadrados x y valor tablero = (((tablero !! coordy) !! coordx) /= valor) && (((tablero !!
(coordy+1)) !! coordx) /= valor) && (((tablero !! (coordy+2)) !! coordx) /= valor) && (((tablero
!! coordy) !! (coordx+1)) /= valor) && (((tablero !! (coordy+1)) !! (coordx+1)) /= valor) &&
(((tablero !! (coordy+2)) !! (coordx+1)) /= valor) && (((tablero !! coordy) !! (coordx+2)) /=
valor) && (((tablero !! (coordy+1)) !! (coordx+2)) /= valor) && (((tablero !! (coordy+2)) !!
(coordx+2)) /= valor)
    where
        coordx = (div x 3) * 3
        coordy = (div y 3) * 3
```

```
m_fila :: [Int] -> Int -> Bool
m_fila [] _ = True
m_fila (x:xs) valor = (x /= valor) && (m_fila xs valor)
```

```
m_columna :: Int -> Int -> [[Int]] -> Bool
m_columna _ _ [] = True
m_columna x valor (y:ys) = ((y !! x) /= valor) && (m_columna x valor ys)
```

```
movimiento_valido :: Int -> Int -> Int -> [[Int]] -> Bool
movimiento_valido x y valor tablero = (m_cuadrados x y valor tablero) && ( m_fila (tablero !!
y) valor ) && (m_columna x valor tablero) && (valor /= 0) && ((tablero !!y) !! x == 0)
```

```

ganaste :: [[Int]] -> Bool
ganaste [] = True
ganaste (x:xs) = fila (x) && ganaste (xs)

```

```

juego :: [[Int]] -> IO ()
juego tablero = do
    putStrLn ("Sudoku by Mateito :D")
    imprimir_tablero tablero
    putStrLn ("Ingresa coordenada x:" )
    x <- getLine
    let x1 = (read x)
    putStrLn ("Ingresa coordenada y:" )
    y <- getLine
    let y1 = (read y)
    putStrLn ("Ingresa valor:" )
    valor <- getLine
    let valor1 = read valor

    putStrLn(show x ++ " " ++ y ++ " " ++ valor)

```

```

if(movimiento_valido x1 y1 valor1 tablero) then do
    let nuevo_tablero = movimiento x1 y1 valor1 tablero
    if (ganaste nuevo_tablero) then
        putStrLn ("Ganaste")
    else do
        juego nuevo_tablero
    else do
        putStrLn ("Movimiento no valido :(((((")
        juego tablero

```

```

main :: IO()
main = do
    let tabl = [[7,8,0,3,0,0,6,0,2],
                [4,0,2,0,8,0,0,7,0],
                [1,3,5,6,0,0,4,9,0],
                [0,0,0,0,0,9,0,3,0],
                [3,0,0,8,0,7,0,0,9],
                [0,9,0,5,0,0,0,0,0],
                [0,1,7,0,0,8,9,6,4],
                [0,2,0,0,5,0,8,0,7],
                [6,0,8,0,0,1,0,2,5]]
    juego tabl

```

Sudoku by Mateito :D

7 8 0 3 0 0 6 0 2

4 0 2 0 8 0 0 7 0

1 3 5 6 0 0 4 9 0

0 0 0 0 0 9 0 3 0

3 0 0 8 0 7 0 0 9

0 9 0 5 0 0 0 0 0

0 1 7 0 0 8 9 6 4

0 2 0 0 5 0 8 0 7

6 0 8 0 0 1 0 2 5

Ingresa coordenada x:

1

Ingresa coordenada y:

1

Ingresa valor:

6

"1"16

Sudoku by Mateito :D

7 8 0 3 0 0 6 0 2

4 6 2 0 8 0 0 7 0

1 3 5 6 0 0 4 9 0

0 0 0 0 0 9 0 3 0

3 0 0 8 0 7 0 0 9

0 9 0 5 0 0 0 0 0

0 1 7 0 0 8 9 6 4

0 2 0 0 5 0 8 0 7

6 0 8 0 0 1 0 2 5

Ingresa coordenada x: